

Node.js Interview Questions (4+ Years Experience)

Core Node.js Concepts

- What is Node.js and how is it different from traditional web server frameworks?
- Explain the event-driven architecture in Node.js.
- What is the event loop? How does it work in Node.js?
- What are streams in Node.js? Types and use cases?
- What are buffers in Node.js?
- What is the difference between `process.nextTick()`, `setImmediate()`, and `setTimeout()`?
- What is the purpose of the `cluster` module in Node.js?

Modules & NPM

- Explain CommonJS vs ES6 Modules in Node.js.
- How does Node.js resolve module imports?
- What is `package-lock.json` vs `package.json`?
- What are `peerDependencies`, `devDependencies`, and `optionalDependencies`?
- What is the difference between a global module and a local module?

File System & Path Handling

- How do you read and write files in Node.js asynchronously?
- Explain the difference between `fs.readFile` and `fs.createReadStream`.
- How do you watch for file changes using `fs.watch()`?

Networking & HTTP

- How do you create a basic HTTP server in Node.js?
- Explain the concept of middleware. How do you write custom middleware?
- What are the benefits of using Express.js over native HTTP?

- How does Node.js handle concurrent connections?
- How do you implement CORS in an Express app?

Express.js (and similar frameworks)

- How does routing work in Express?
- What is middleware chaining? What are the different types of middleware?
- How do you handle errors in Express?
- How can you structure a large Express app?
- How to secure Express apps (helmet, rate limiting, etc.)?

Security

- What are common security risks in Node.js apps (e.g., XSS, CSRF, injection)?
- How do you prevent SQL injection / NoSQL injection?
- How do you manage environment variables securely?
- What is CORS and how do you configure it in Node.js?
- How do you implement JWT authentication?

Database Integration

- How do you connect Node.js to MongoDB / PostgreSQL / MySQL?
- Difference between using ORM (like Sequelize/TypeORM) and raw queries?
- What are connection pooling and its benefits?
- How do you handle transactions in Node.js?

Asynchronous Programming

- Explain callbacks vs promises vs async/await.
- What are async iterators and generators in Node.js?
- How to handle errors in asynchronous code?

- What is the `util.promisify` function?

Testing & Debugging

- What testing frameworks do you use (Mocha, Jest, Supertest)?
- How do you write unit and integration tests in Node.js?
- How to mock dependencies in Node.js tests?
- How do you debug a Node.js app (VS Code, Chrome DevTools, etc.)?

Architecture & Design

- How do you design scalable REST APIs in Node.js?
- What is a microservice? Have you implemented one with Node.js?
- What are common design patterns used in Node.js (e.g., Singleton, Factory)?
- How do you structure Node.js applications for maintainability?

Performance & Optimization

- How do you improve performance in a Node.js application?
- What is the role of caching (Redis, in-memory) in Node.js?
- What are memory leaks? How do you identify and fix them?
- How do you monitor a Node.js application in production?

DevOps / Deployment

- How do you deploy a Node.js app (Docker, PM2, etc.)?
- What are your steps for CI/CD for a Node.js app?
- How do you manage environment configs across environments?
- How do you handle logs and errors in production?

Situational & System Design Questions

- Design a REST API for a Todo app with authentication.
- How would you build a real-time chat app in Node.js?
- How would you scale a Node.js application to handle 10k+ concurrent users?
- Whats your approach to handling rate limiting and abuse?